ECE448 Spring 2025

Lesson 02

Link Budget

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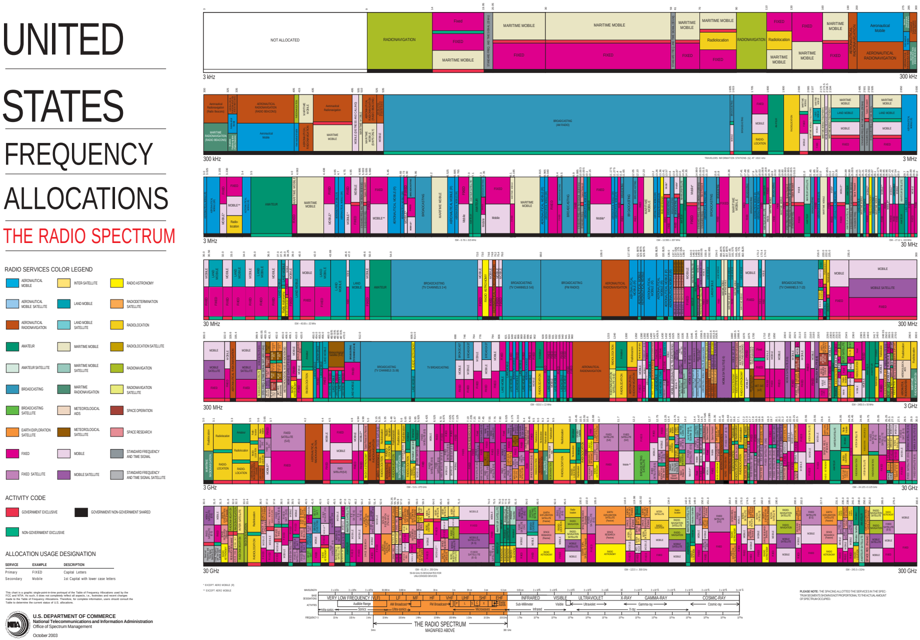
Lesson 2 Learning Objectives

1.1 Identify key bands within the FCC bandwidth allocations.

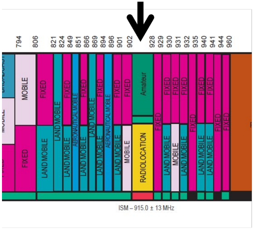
1.2 Identify sources of loss in a link budget and perform a link budget analysis.

1.3 Identify types and sources of noise and how to incorporate them into a link budget.

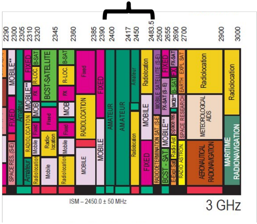

UNITED
STATES
FREQUENCY
ALLOCATIONS
THE RADIO SPECTRUM
RADIO SERVICES COLOR LEGEND
AERONAUTICAL MOBILE
INTER-SATELLITE
RADIO ASTRONOMY
AERONAUTICAL MOBILE SATELLITE
LAND MOBILE
RADIODETERMINATION SATELLITE
AERONAUTICAL RADIONAVIGATION
LAND MOBILE SATELLITE
RADIOLOCATION
AMATEUR
MARITIME MOBILE
RADIOLOCATION SATELLITE
AMATEUR SATELLITE
MARITIME MOBILE SATELLITE
RADIONAVIGATION
BROADCASTING
MARITIME RADIONAVIGATION
RADIONAVIGATION SATELLITE
BROADCASTING SATELLITE
METEOROLOGICAL AIDS
SPACE OPERATION
EARTH EXPLORATION SATELLITE
METEOROLOGICAL SATELLITE
SPACE RESEARCH
FIXED
MOBILE
STANDARD FREQUENCY AND TIME SIGNAL
FIXED SATELLITE
MOBILE SATELLITE
STANDARD FREQUENCY AND TIME SIGNAL SATELLITE
ACTIVITY CODE
GOVERNMENT EXCLUSIVE
GOVERNMENT/ NON-GOVERNMENT SHARED
NON-GOVERNMENT EXCLUSIVE
ALLOCATION USAGE DESIGNATION
SERVICE
EXAMPLE
DESCRIPTION
Primary
FIXED
Capital Letters
Secondary
Mobile
1st Capital with lower case letters
3 kHz
300 kHz
3 MHz
30 MHz
300 MHz
3 GHz
30 GHz
300 GHz
THE RADIO SPECTRUM
MAGNIFIED ABOVE
U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management
October 2003

FCC Allocations: ISM Bands

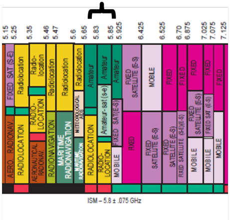
902 – 928 MHz
User: Piccolo, 3DR
BW: 26MHz



2.4 – 2.4835 GHz
Users: WiFi, RC radios
BW: 83.5MHz

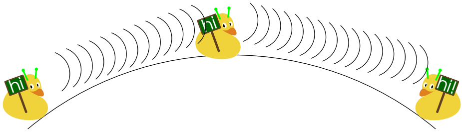


5.725 – 5.875 GHz
Users: WiFi, video
BW: 150MHz



Wireless Comms - Factors

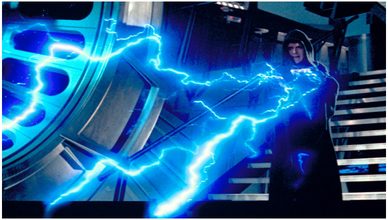
LOS : always check first!



Wireless Comms - Factors

LOS : always check first!

Power transmitted

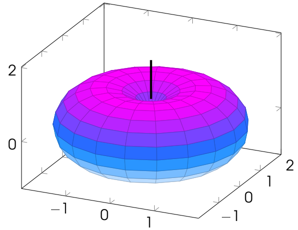


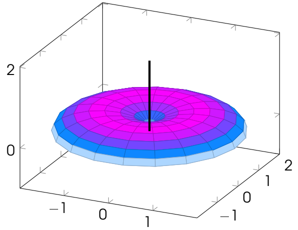
Wireless Comms - Factors

LOS : always check first!

Power transmitted

Gain of Tx/Rx antenna





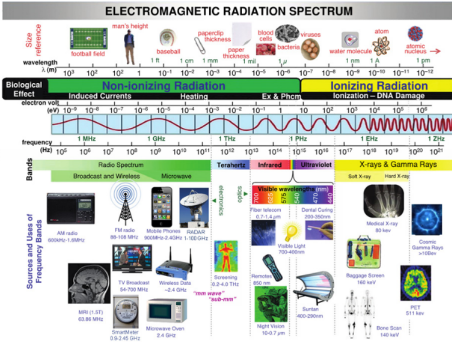
Wireless Comms - Factors

LOS : always check first!

Power transmitted

Gain of Tx/Rx antenna

Wavelength (frequency)



Wireless Comms - Factors

- LOS : always check first!
- Power transmitted
- Gain of Tx/Rx antenna
- Wavelength (frequency)
- Distance of communication

Wireless Comms - Factors

LOS : always check first!

Power transmitted

Gain of Tx/Rx antenna

Wavelength (frequency)

Distance of communication

Friis Equation

Friis Equation

$$P_{\mathrm{R}} = \underline{\hspace{4cm}}$$

P_{R}

P_{R} = power collected by the receiving antenna (W)

Friis Equation

$$P_{\text{R}} = \frac{P_{\text{T}}}{\qquad\qquad}$$

P_{R}

P_{R} = power collected by the receiving antenna (W)

P_{T} = power sent by the transmitting antenna (W)

Friis Equation

$$P_R = \frac{P_T G_T}{\quad}$$

P_R

G_T

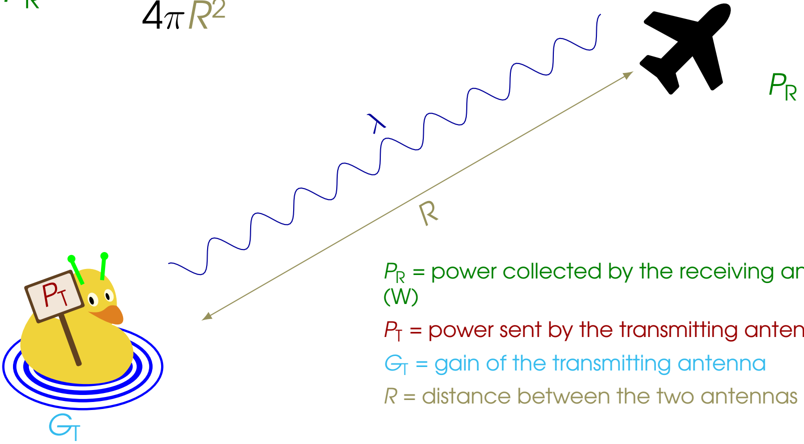
P_R = power collected by the receiving antenna (W)

P_T = power sent by the transmitting antenna (W)

G_T = gain of the transmitting antenna

Friis Equation

$$P_R = \frac{P_T G_T}{4\pi R^2}$$



P_R = power collected by the receiving antenna (W)

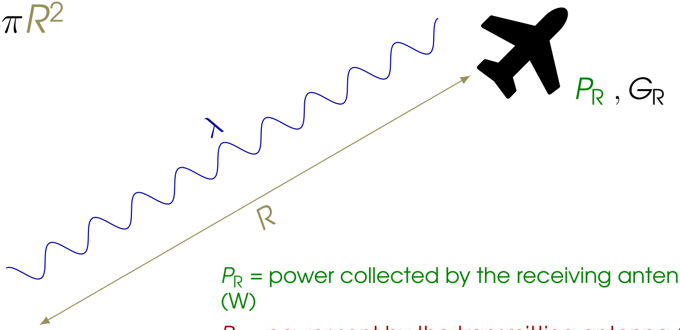
P_T = power sent by the transmitting antenna (W)

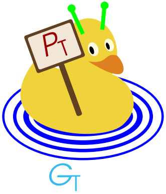
G_T = gain of the transmitting antenna

R = distance between the two antennas

Friis Equation

$$P_R = \frac{P_T G_T G_R A_e}{4\pi R^2}$$





P_R = power collected by the receiving antenna (W)

P_T = power sent by the transmitting antenna (W)

G_T = gain of the transmitting antenna

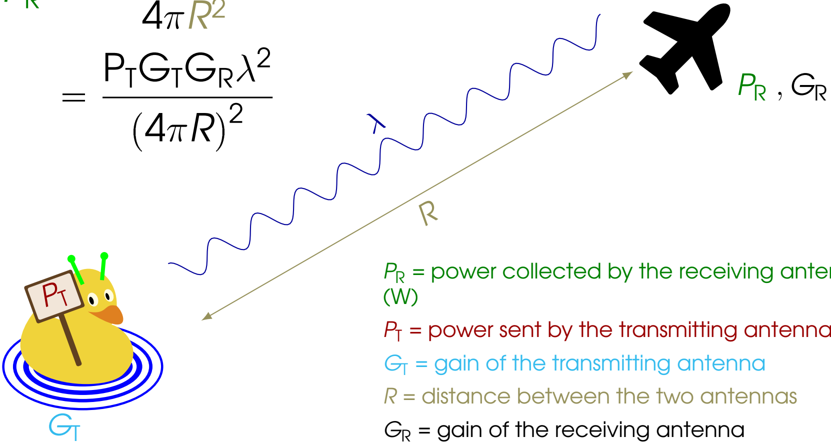
R = distance between the two antennas

G_R = gain of the receiving antenna

Friis Equation

$$P_R = \frac{P_T G_T G_R A_e}{4\pi R^2}$$

$$= \frac{P_T G_T G_R \lambda^2}{(4\pi R)^2}$$



P_R = power collected by the receiving antenna (W)

P_T = power sent by the transmitting antenna (W)

G_T = gain of the transmitting antenna

R = distance between the two antennas

G_R = gain of the receiving antenna

A_e = effective receiving aperture

λ = wavelength of the signal (m)

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Friis & Decibels

- Friis equation (algebraic form)

$$P_R = \frac{P_T G_T G_R \lambda^2}{(4\pi R)^2}$$

- Friis equation (dB form)

$$\begin{aligned} P_{R,dB} &= P_{T,dB} + G_T + G_R + 20\log_{10}(\lambda) - 20\log_{10}(4\pi) - 20\log_{10}(R) \\ &= P_{T,dB} + G_T + G_R + 20\log_{10}(c) - 20\log_{10}(f) \\ &\qquad - 20\log_{10}(4\pi) - 20\log_{10}(R) \\ &= P_{T,dB} + G_T + G_R + 147.5\text{ dB} - 2f_{\text{dB}} - 20\log_{10}(R) \end{aligned}$$

Friis (dB form)

$$P_{R,dB} = P_{T,dB} + G_T + G_R - L_P$$

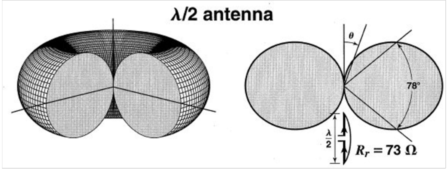
$$\Longrightarrow L_P = \text{Path Loss} = 2f_{\text{dB }(Hz)} + 2R_{\text{dB (meter)}} - 147.5\text{ dB}$$

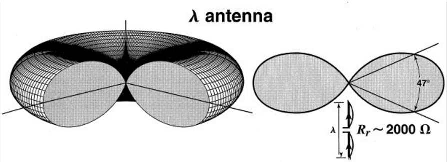
FRIIS & DECIBELS: EXAMPLE

- For the given system, convert values to dB

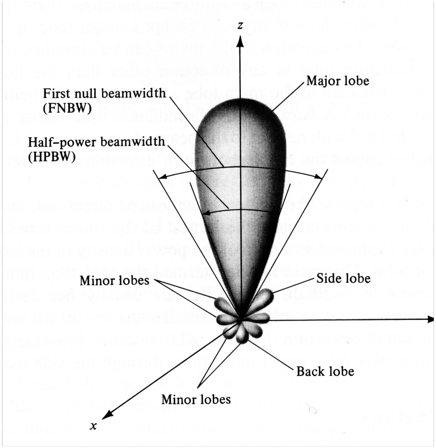
Parameter	**Linear**	**dB**
P_t	120W	
G_t	1201.2	
$G_r t$	281.5	
f	3.7 GHz	
R	80 km	

Antenna Considerations





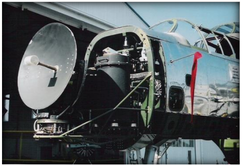
Source: Antennas: For All Applications, p. 179, Kraus, 2002.



Source: Antennas Theory, p. 31, Balanis, 1997.

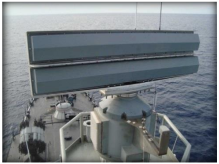
Antennas

Antennas transfer energy contained in circuits (voltages/currents) to electromagnetic radiation (E and H fields)

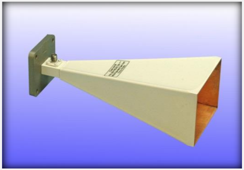


Early Dish Antenna (WWII)

Large Parabolic Dish



Slotted Waveguide



Horn

Passive Phased Array

Active Phased Array

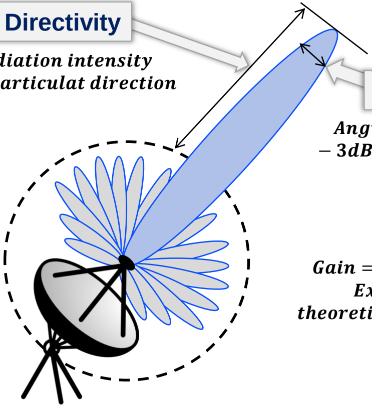
Antenna Parameters

Directivity

Radiation intensity in a particulat direction

Effective Aperture

How much power is transferred from an ideal plane wave



Beamwidth

Angular seperation of $-3dB$ *points of main lobe*

Gain

$Gain = Directivity \times effeceincy$

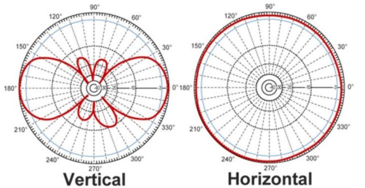
Expressed relative to a theoretical isotropic radiator (dBi)

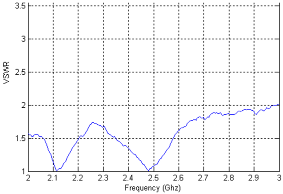
Antenna Design Objectives:

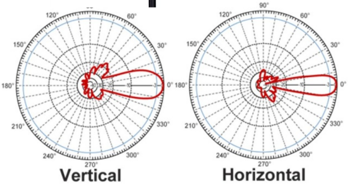
- Accentuate radiation in desired directions, suppress in others
- Achieve maximum *circuit* ⇔ *space* power transfer

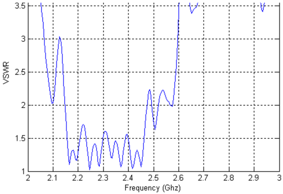
Antenna Considerations: Gain Pattern

L-COM 4dBi Omnidirectional Antenna

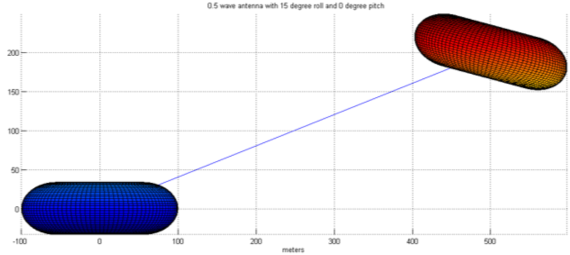






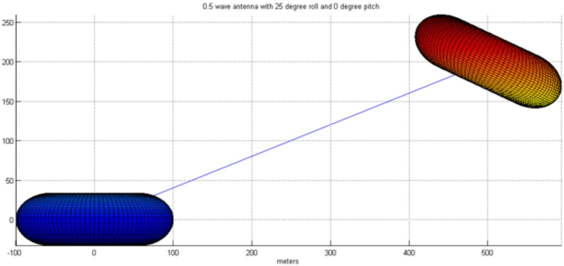


Antenna Considerations: Orientation



$\longleftarrow 15^\circ$

$25^\circ \longrightarrow$



Noise

"Hey Bear, can you hear me now?"

Noise

SNR

$$SNR = \frac{P_{sig}}{P_{noise}}$$

Noise Figure

$$NF = \frac{SNR_{in}}{SNR_{out}} \longrightarrow \text{Typical 2-8dB}$$

System Noise Figure

For a system of cascaded components:

$$F = 1 + \frac{P_{int}}{G_a kTBW}$$

Noise Sources

Shot Noise

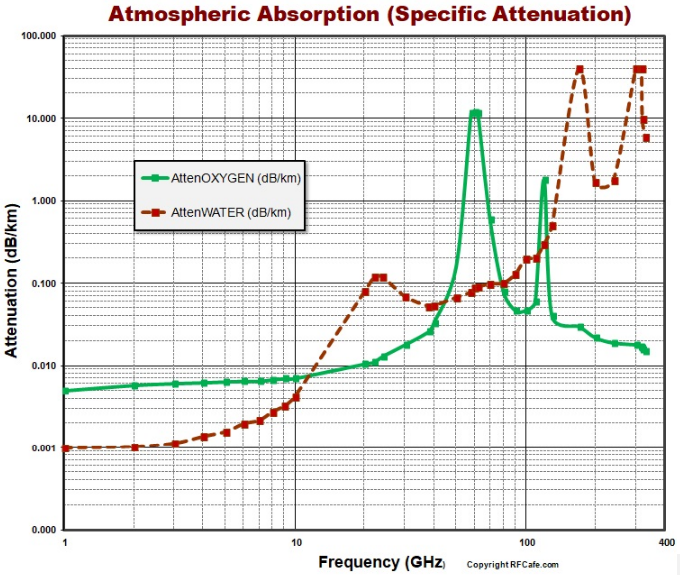
Arises from the flow of electrons in semiconductor devices:

$$i_{\mathrm{rms}}^2 \approx 2kTBg_0 \implies P = kTBW$$
$$\implies T = \text{ Noise temperature}$$

Other Sources

- Generation-recombination noise
- Temperature fluctuation noise
- Quantum noise (hf)

ATMOSPHERIC



Link Budget: Example

Goal is to have $\approx$ 10 dB of margin

Parameter	Value
P_t	630mW
$P_{rx,min}$	-93dBm
G_{tx}	4dBi
G_{rx}	1dBi
f	2.45 GHz
R_{max}	3 km
Max bank angle (HPBW)	20°
Insertion loss (radio-antenna)	1.5dB

Link Budget: Homework

Task: Conduct a link budget analysis for a consumer-grade GPS system, based on realistic, commonly available devices. You will need to research the appropriate parameters, make (and document) reasonable assumptions, and produce a report detailing your analysis.
Due date: 17 Jan 2025 (COB)